

PREBOARDING BATCH - PROJECT CASE STUDY

TELECOM CUSTOMER & SUBSCRIPTION MANAGEMENT SYSTEM

Telecom BSS – Customer, Plan, Subscription & Billing

1. Business Scenario

A telecommunications service provider offers mobile voice, SMS, data, international roaming and value-added services to millions of customers.

The company wants to develop a centralized **Telecom Customer & Subscription Management System (TCSMS)** to manage:

- Customer onboarding
- Customer profiles
- Mobile numbers
- SIM cards
- Tariff plans
- Subscriptions
- Usage
- Billing
- Payments
- Complaints
- Notifications

The system must support two major roles:

- **Customer**
- **Telecom Administrator**

2. Application Login

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TELECOM CUSTOMER MANAGEMENT SYSTEM

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1. Customer Login
2. New Customer Registration
3. Administrator Login
4. Forgot Password
5. Exit

Authentication should include:

- Username/password
- CAPTCHA
- Maximum 3 failed attempts
- Temporary account lock
- OTP-based password recovery
- Login history
- Last login timestamp
- Logout

3. Customer Registration

The customer should provide:

customerId

customerNumber

firstName

lastName

dateOfBirth

email

mobileNumber

address

city

country

username

passwordHash

registrationDate

accountStatus

Example:

Customer No.	Name	City	Status
CUST100245	Arjun Mehta	Mumbai	ACTIVE
CUST100378	Sarah Wilson	London	ACTIVE
CUST100412	Omar Hassan	Riyadh	ACTIVE

Business Rules

- Email must be unique.
- Customer mobile number must be unique.
- Password must satisfy defined complexity rules.
- Customer must satisfy minimum age requirements.
- Mandatory fields must be validated.

4. Mobile & SIM Management

A customer can have one or more mobile connections.

MobileSubscription

subscriptionId

subscriptionNumber

customerId

mobileNumber

simNumber

simType

activationDate

subscriptionType

status

SIM types:

PHYSICAL_SIM

ESIM

Subscription types:

PREPAID

POSTPAID

Example:

Mobile Number : +91-9876543210

SIM : ESIM

Type : POSTPAID

Status : ACTIVE

5. Telecom Plan Catalogue

The administrator maintains tariff plans.

TelecomPlan

planId

planCode

planName

planType

monthlyRental

dataAllowanceGB

voiceMinutes

smsAllowance

validityDays

internationalRoaming

status

Example:

Plan	Type	Data	Voice	Monthly Rental
PLAN-101	5G Premium	100 GB	Unlimited	₹999
PLAN-102	5G Standard	50 GB	1500 min	₹699
PLAN-103	Business Pro	200 GB	Unlimited	₹1,499
PLAN-104	Basic	20 GB	500 min	₹399

Customers can:

- Search plans
- Filter by price
- Filter by data allowance
- Sort by price
- Compare plans

Use **Lambda**, **Comparator** and **Stream API**.

6. Subscription Management

Customers can:

- Subscribe to a plan
- Upgrade plan
- Downgrade plan
- Change prepaid/postpaid plan where permitted
- Activate/deactivate add-on services

SubscriptionHistory

historyId

subscriptionId

oldPlanId

newPlanId

changeDate

changeReason

changedBy

Business rules:

- Inactive plans cannot be selected.
- A customer cannot subscribe to the same plan twice.
- Plan changes must be recorded in history.
- Certain plans may have upgrade restrictions.

7. Usage Management

The system should maintain telecom usage.

UsageRecord

usageId

subscriptionId

usageDate

usageType

quantity

unit

charge

Usage types:

VOICE

SMS

DATA

ROAMING

Example:

DATA → 2.5 GB

VOICE → 120 Minutes

SMS → 45

ROAMING → 350 MB

Use Java 8 Stream API to calculate:

- Total data usage
- Total voice usage
- Total SMS
- Monthly usage
- Usage by type
- Highest-consuming customers

8. Billing

Generate monthly bills.

Bill

billId

billNumber

subscriptionId

billingMonth

planRental

usageCharges

taxAmount

discount

totalAmount

dueDate

billStatus

Example:

Bill Number : INV-2026-08-10245

Customer : CUST100245

Plan Rental : ₹699

Usage Charges : ₹320

Tax : ₹183

Discount : ₹50

Total : ₹1,152

Due Date : 20-Aug-2026

9. Payment Processing

Payment

paymentId

transactionReference

billId

customerId

amount

paymentMode

paymentDate

paymentStatus

Payment modes:

UPI

CARD

NET_BANKING

BANK_TRANSFER

Payment must use a JDBC transaction:

Validate Bill

↓

Validate Amount

↓

Create Payment

↓

Update Bill Status

↓

Create Audit Record

↓

COMMIT

Failure → ROLLBACK.

Duplicate payment must not be allowed.

10. Administrator Module

Administrator can:

- Manage customers
- Manage plans
- Manage SIMs
- Manage subscriptions
- View usage
- Generate bills
- View payments
- Suspend/reactivate subscriptions
- Manage complaints
- Generate reports

11. Customer Complaints

Complaint

complaintId

complaintNumber

customerId

subscriptionId

category

description

priority

createdDate

status

resolution

Categories:

BILLING

NETWORK

SIM

PLAN

PAYMENT

OTHER

12. Java 8 Requirements

Use Stream API for:

- Highest-usage customers
- Customers by city
- Plans within a price range
- Most subscribed plans
- Monthly revenue
- Usage by type
- Customers with unpaid bills
- Average monthly revenue per customer

Use:

Predicate

Function

Consumer

Supplier

Comparator

Optional

Collectors.groupingBy()

Collectors.summarizingDouble()

13. Multithreading

Implement background processes:

BillingScheduler

Generates monthly bills.

PaymentNotificationService

Processes payment notifications.

UsageProcessor

Processes usage records.

AccountMonitor

Identifies suspended/overdue accounts.

Use:

ExecutorService

ScheduledExecutorService

Runnable

Callable

Future

Synchronization

14. Suggested Classes

Customer

Administrator

MobileSubscription

SIMCard

TelecomPlan

SubscriptionHistory

UsageRecord

Bill

Payment

Complaint

Notification

AuditLog

LoginHistory

AuthenticationService

CustomerService

SubscriptionService

PlanService

BillingService

PaymentService

ComplaintService

ReportService

CustomerDAO

SubscriptionDAO

PlanDAO

UsageDAO

BillingDAO

PaymentDAO

ComplaintDAO

CaptchaGenerator

OTPService

PasswordUtil

DBConnection

ReportGenerator

COMMON DATABASE REQUIREMENT

Each project must have a properly normalized relational database.

Participants should demonstrate:

Tables

- Primary Keys
- Foreign Keys
- Unique Constraints
- NOT NULL
- Appropriate indexes
- Status columns
- Created/Updated timestamps

SQL

At minimum:

SELECT

INSERT

UPDATE

DELETE

JOIN

LEFT JOIN

GROUP BY

HAVING

ORDER BY

SUBQUERY

CASE

AGGREGATE FUNCTIONS

DATE FUNCTIONS

Where appropriate, participants may implement:

Views

Stored Procedures

Functions

Triggers

Indexes

COMMON JAVA ARCHITECTURE

A suitable package structure is:

com.amdocs.telecom

|

|— controller

|

|— service

| └─ impl

|

|— dao

| └─ impl

|

|— model

|

|— dto

|

|— exception

|

|— validation

|

|— security

|

|— scheduler

|

|— report

|

|— util

|

└─ main

Participants should demonstrate:

Core Java

- Classes and Objects
- Encapsulation
- Inheritance
- Abstraction
- Polymorphism
- Interfaces
- Enums
- Generics
- Collections

Java 8

- Lambda
- Functional Interfaces
- Stream API
- Optional
- Method References
- Default/Static Interface Methods

JDBC

- Connection
- PreparedStatement
- ResultSet
- Batch Processing
- Transactions
- Commit
- Rollback
- Savepoint

Multithreading

- Thread/Runnable
- ExecutorService
- Callable/Future
- ScheduledExecutorService
- Synchronization
- BlockingQueue

Security

- CAPTCHA
- Password hashing
- OTP
- Account locking
- Role-based authorization
- PreparedStatement

File Handling

- CSV export
- Text reports
- Configuration files
- Logging

Design Patterns

Minimum three, such as:

DAO

Factory

Strategy

Singleton

Observer